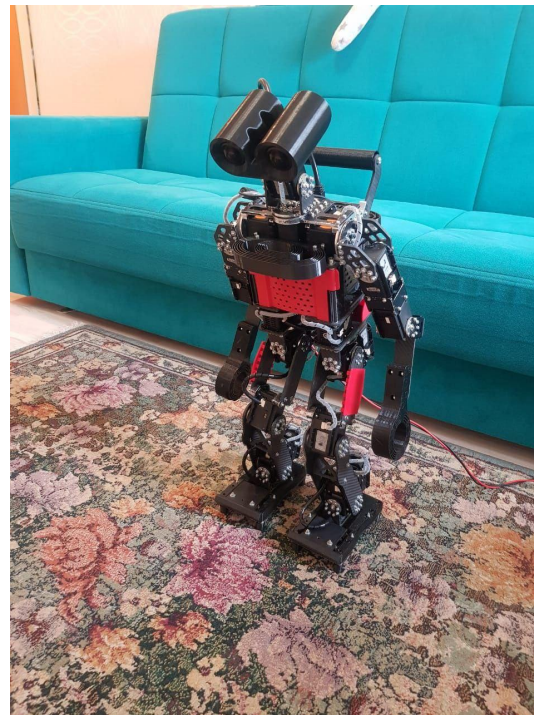


Legged locomotion

Ilya Osokin, 2026

What a robot is actually?

- Sensors
- World model
- Localization (state estimation)
- Planning
- Motion



Walking

- Bipedal/quadrupedal
- Common among animals
- Challenging for control engineers



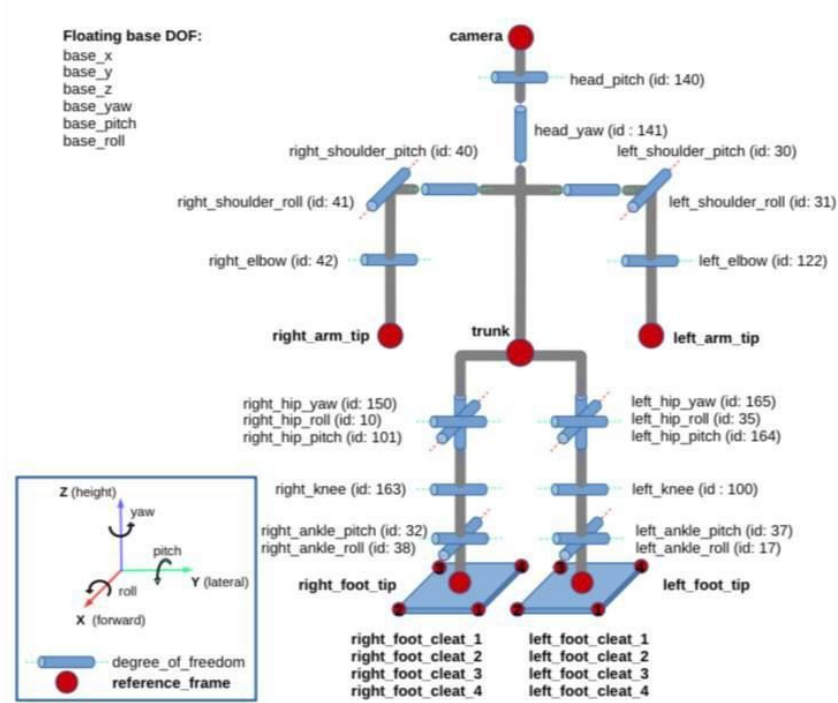
SAHR



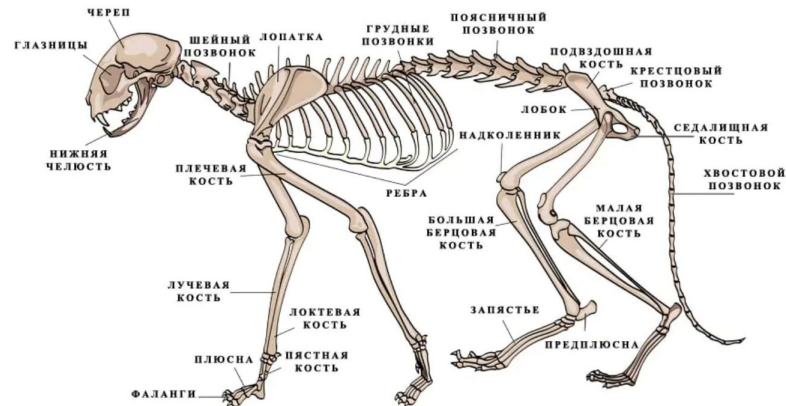
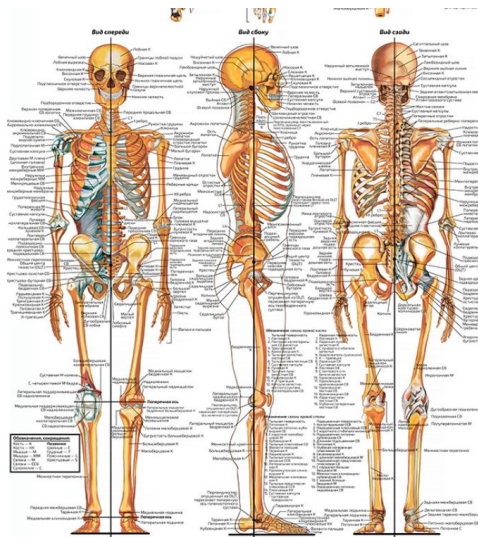
Сервомоторы	
10 x servos Robotis Dynamixel MX64 10 x servos Robotis Dynamixel MX106	8 x servos Robotis Dynamixel MX64 12 x servos Robotis Dynamixel MX106
Зрение	
1 x камера FLIR Blackfly S GigE 1440x1080 78fps	2 x камеры FLIR Blackfly S GigE 1440x1080 78fps
Сенсоры	
4-канальные датчики давления AD7124 ADC converter board with Dynamixel v2.0 interface (STM32 based)	
Bosch BHI260AB IMU board with Dynamixel v2.0 interface controller (STM32 based)	
Бортовой ПК	
Intel NUC	
Intel Core i5 4x1.60GHz	Intel Core i7 6x1.10GHz
8 GB RAM	16 GB RAM
256 GB SSD	512 GB SSD
Питание	
Battery 3300mAh 14.8V	
Материалы	
карбон, алюминий, пластик (3D печать)	
Размеры	
73 x 26 x 27 cm	70 x 26 x 27 cm
Вес	
6.8 kg	7.2 kg



Degrees of freedom

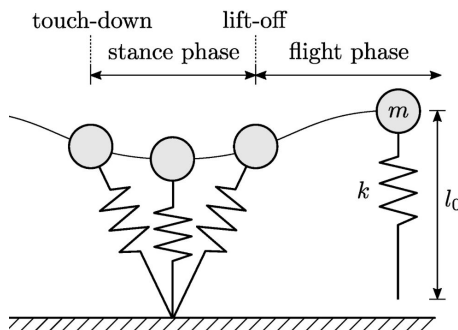
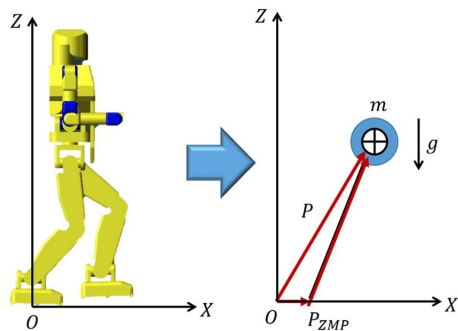


Kinematic schemes



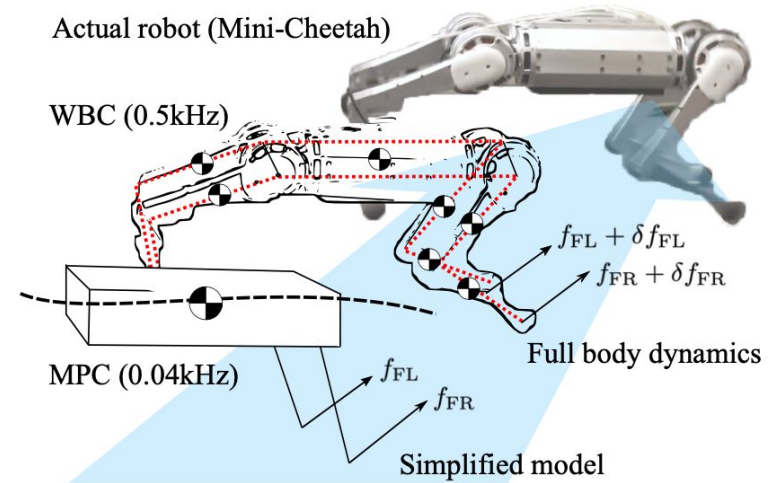
Walking algorithms

- Kinematic walking - slow, safe, unnatural
- Zero Moment Point (ZMP)
- Inverted pendulum, SLIP

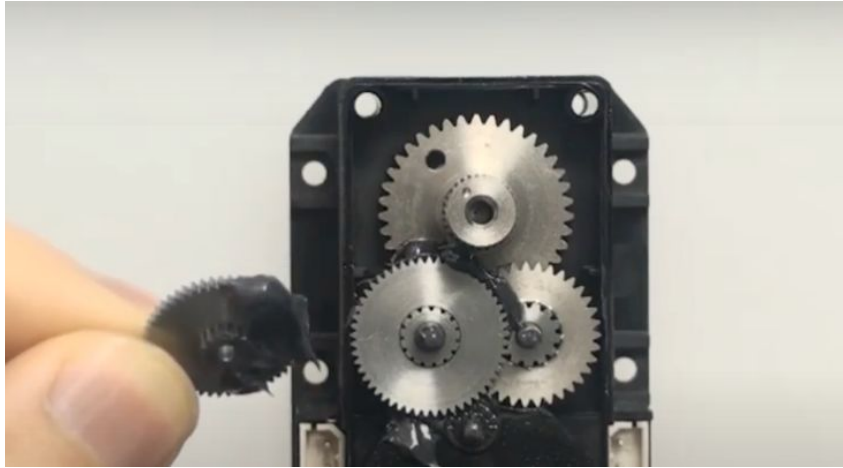


Control levels

- Dijkstra and alike; 1 Hz
- Foot placement, Center of Mass; 10 Hz
- Whole Body, MPC 100 Hz
- Impedance control, individual servos; 1kHz
- Hardware; ∞ Hz

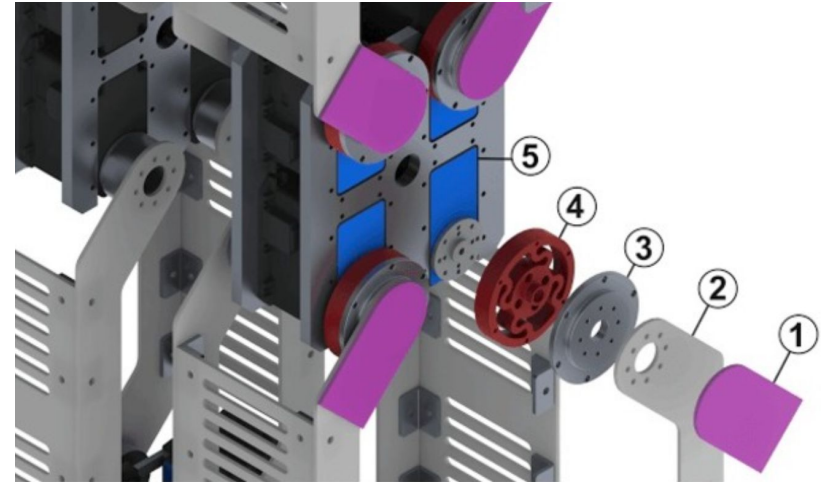


Servomotors



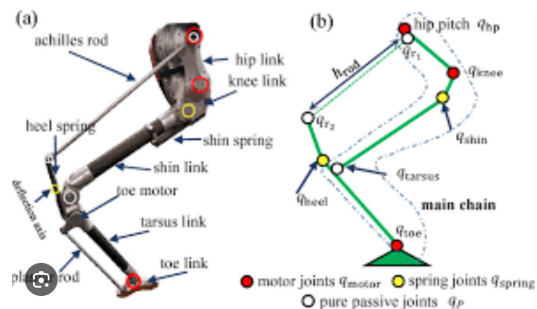
Serial elastic elements

- Low-pass
- Dampers the hitting impact
- Complicates the control
- Complicates the mechanical design



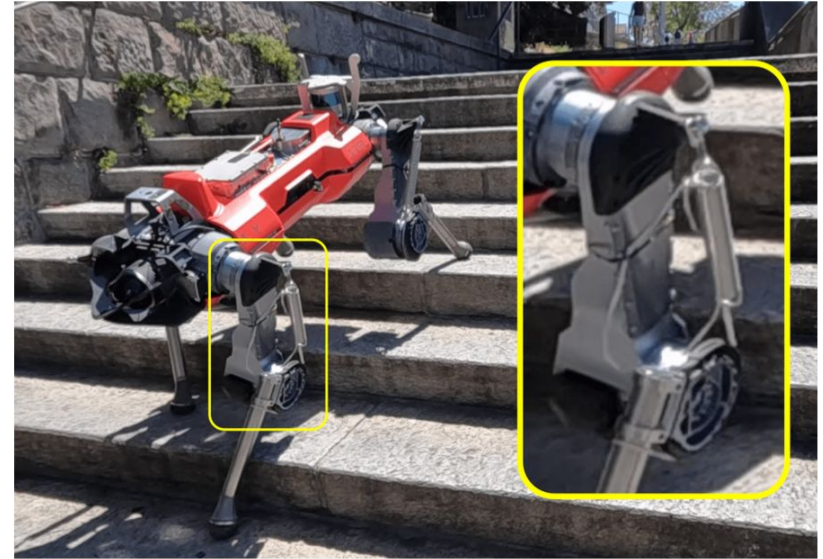
Cassie

- Bird-like legs
- Serial elastics
- 5 km on one charge
- World record in bipedal 100 m



Parallel elastic elements

- Recurring load compensation
- Complicates the mechanical design
- Could (potentially) be adaptive

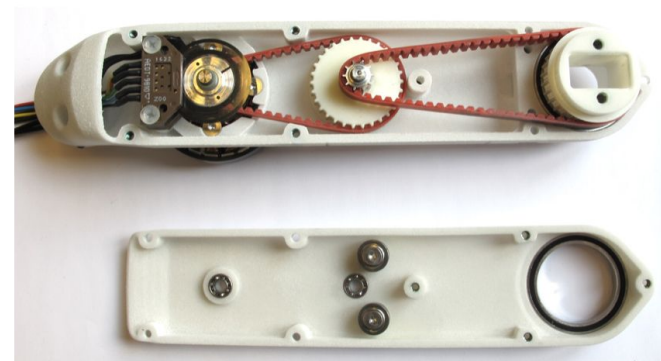
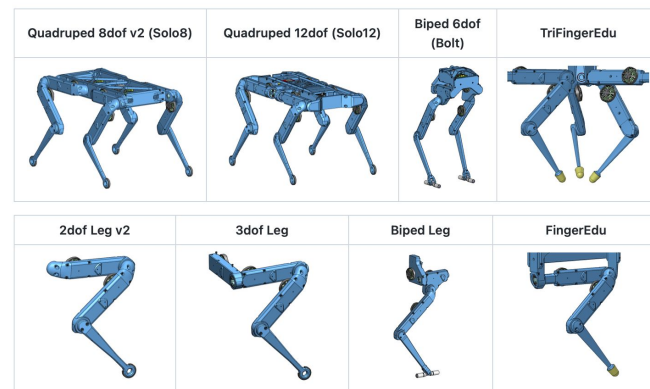


Simulation, domain randomization

- Varying parameters of friction, masses, ...
- Complicating “dataset”



Cheaper and easier



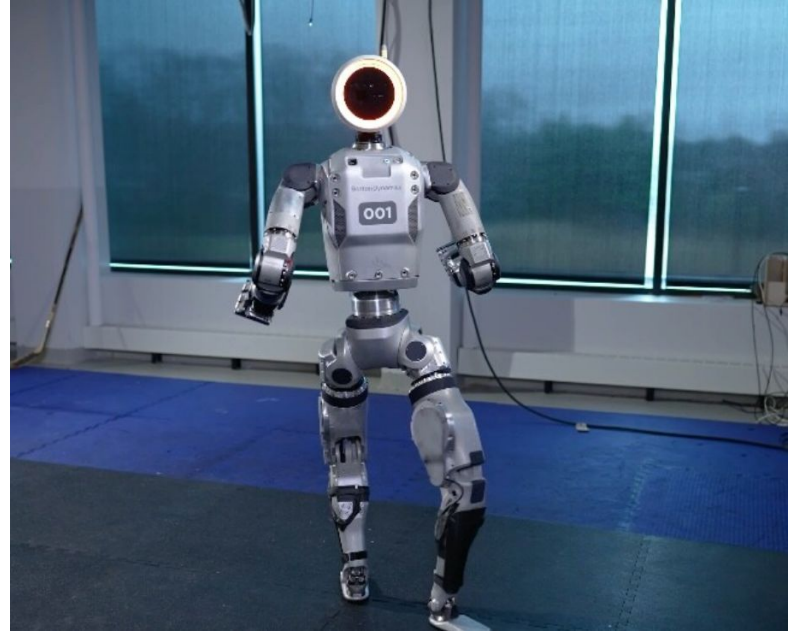
Recent trends

- Joints in torso
- Linkage-based feet actuation
- High-torque hip and knee pitch
- Less strict joint limits



Fourier GR-1

How far are we?



Actual problems

- Manipulation (multi-contact point)
- Flight phase
- Feedback density
- Awareness (understanding)
- Rigidity and fragility of the actuators

Problems in educational robotics

- High entrance barrier (including teachers)
- Absence of a straightforward trajectory
- Common hardware is fundamentally different to the industry (Lego)
- Sanctions



???



Contacts

